
AWS Global Accelerator – Complete Overview

✅ What is AWS Global Accelerator?

AWS Global Accelerator is a **networking service** that improves the **availability, performance, and global reach** of your applications by directing traffic through the AWS **global network infrastructure**.

Instead of relying on the **public internet**, it provides a **dedicated path** with **static IP addresses**, low latency, and **automatic failover** for highly available apps.

🔧 Key Features

Feature	Description
Static IP Addresses	Get 2 anycast IPs for your accelerator, which remain constant regardless of endpoint changes.
Global Reach	Optimizes the path to your app using the AWS global network backbone.
Automatic Health Checks	Detects unhealthy endpoints and reroutes traffic.
Traffic Distribution	Based on **geolocation**, **weights**, or **failover**.
DDoS Protection	Integrated with **AWS Shield** for security.
Elastic Load Balancer (ELB) & EC2 Support	Supports ALB, NLB, and EC2 as endpoints.
Client IP Preservation	Preserves the original client IP address.

📌 When to Use Global Accelerator

- Global **web applications** with users from multiple countries.
- **Latency-sensitive** applications like VoIP, gaming, or trading platforms.
- **Failover-critical** apps that need auto-failover between regions.
- Centralized endpoints for APIs or SaaS apps.

⚙️ Components of Global Accelerator

1. Accelerator

- The top-level resource.
- Provides **2 static IPs** for global traffic.
- Routes requests to the closest endpoint using the AWS backbone.

2. Listener

- Listens on **TCP/UDP ports** (e.g., 80, 443).
- Attached to the accelerator.
- Defines how to process incoming requests.

3. Endpoint Groups

- A group of endpoints in a specific AWS Region.

- You can define:
    - **Traffic dial** (0–100%) to shift traffic gradually.
    - **Health checks** for routing.

4. Endpoints

- Can be:
    - **Elastic IP**
    - **ALB (Application Load Balancer)**
    - **NLB (Network Load Balancer)**
    - **EC2 instance**
    - **Elastic Fabric Adapter (EFA)**

🌍 How It Works (Internals)

1. Users connect to the **static IP** of the accelerator.
2. Global Accelerator routes the request to the **nearest edge location** using **Anycast**.
3. Traffic is routed over AWS's **private backbone network** to the **best endpoint**.
4. If an endpoint or region fails, traffic is automatically rerouted.

💡 Benefits

Benefit	Why It Matters
Improved Performance	Reduces latency by routing over AWS backbone.
High Availability	Built-in failover and health checks.
Easy Maintenance	Endpoint changes don't affect client IPs.
Security	Supports AWS Shield Standard.

💰 Pricing (as of 2025)

- **$0.025/hour** per accelerator.
- **$0.015/GB** data transfer over accelerator.

> ⚠️ Prices vary by region and traffic type. Always check AWS Pricing.

🧪 Example Use Case

You have users in **Europe and Asia**, and your backend is hosted in **us-east-1** and **ap-southeast-1**. With Global Accelerator:

- Users in Europe reach `us-east-1` endpoint via AWS backbone.
- Users in Asia are routed to `ap-southeast-1`.
- Failover and health checks ensure high availability.

📚 **Summary**

Term	Explanation
Accelerator	Entry point with static IPs.
Listener	Accepts incoming traffic.
Endpoint Group	Group of regional endpoints.
Endpoint	EC2/ALB/NLB resources.
Traffic Dial	Controls traffic percentage.

❓ Do You Still Need a CDN (like Amazon CloudFront) if You Use AWS Global Accelerator?

✅ **Short answer:**
Yes, Global Accelerator and CloudFront serve *different purposes*, and they complement each other. In many architectures, you may use both together.

🆚 Global Accelerator vs CloudFront

Feature	AWS Global Accelerator	Amazon CloudFront (CDN)
Use Case	Optimize performance for **dynamic, non-cacheable** content	Distribute and cache **static and dynamic** content

Primary Goal	Improve **latency** and **availability** using AWS backbone	**Cache content** close to users at edge locations
IP Address Type	Provides **2 static IPs (anycast)** for global access	No static IP; uses **domain name (DNS)**
Caching	❌ No caching	✅ Caching supported
Protocol Support	TCP/UDP	HTTP/HTTPS
Best For	APIs, Gaming, VoIP, Real-time apps, cross-region failover	Images, videos, HTML, JS, CSS (static content)
DDoS Protection	Integrated with AWS Shield	Also integrated with AWS Shield
Edge Locations	Routes to nearest AWS Region via backbone	Caches at edge (200+ POPs globally)
Example	Routing API calls to us-west-2 or ap-southeast-1	Serving a JS file globally with low latency

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🧩 Use Them Together

You **can and often should** use **both**:

- 🔄 **CloudFront** for **static content** (images, JS, CSS, media)
- 🌍 **Global Accelerator** for **dynamic requests or APIs**

💡 Example Architecture:

sql

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```
User ---> CloudFront (Static Content)
```

```
     \
      ---> Global Accelerator ---> ALB/EC2 in closest region
(Dynamic API)
```

🔚 Conclusion

Global Accelerator ≠ CDN.

- Use **CloudFront** when caching is beneficial.
- Use **Global Accelerator** to optimize latency and availability for dynamic content.
- Together, they offer **global reach**, **low latency**, **high availability**, and **resilience**.

Would you like a **comparison architecture diagram** or **combined lab** using both?

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